ARTICLE

Relative fat mass as an estimator of body fat percentage in Chilean adults

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BACKGROUND/OBJECTIVE: The Relative Fat Mass (RFM) is an alternative index to body mass index (BMI) for estimating whole body fat percentage (BF%). Our aims were to determine the accuracy of the RFM for 1) identifying individuals with elevated BF% and, 2) estimating the BF% compared to Dual-energy X-ray absorptiometry (DXA) in a Chilean adult population.

SUBJECTS/METHODS: Body composition was assessed by DXA in 270 healthy participants (125 women/145 men). Anthropometric measurements were assessed to calculate RFM and BMI. Receiver operating characteristic (ROC) curves were obtained to assess the sensitivity and specificity of both, RFM and BMI. Bland–Altman analysis between BF% measured by DXA vs. predicted BF% derived from RFM was performed to assess validity. Pearson´s correlation coefficients to analyze the association between BMI, RFM and DXA were also calculated.

RESULTS: For RFM, the cut-off for elevated BF% was ≥22.7% for men and ≥32.4% for women and for BMI was ≥24.4 kg/m^2 for men and ≥24.1 kg/m^2 for women. The area under the ROC curve between RFM and BMI was not significantly different in men (0.970 vs. 0.959; $p = 0.420$) and women (0.946 vs. 0.942, $p = 0.750$). The Bland–Altman analysis showed that the estimation bias is more pronounced in men than in women.

CONCLUSION: RFM is an accurate tool for identifying individuals with elevated BF%, although it was not as accurate as DXA for estimating the BF%. RFM may be an alternative method useful in primary care to select individuals for lifestyle counseling and in research to select patients for epidemiological studies.

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INTRODUCTION

The prevalence of obesity has increased two-fold since 1980 and it has become a worldwide health problem. It is estimated that one third of the world's population live with overweight or obesity [1]. In Chile, 74% of adults have some degree of excessive body weight [2] and worldwide, Chile is the one of the countries with the highest prevalence of obesity [3]. The increasing prevalence of obesity is associated with health impairments, and reduced quality of life, involving higher health costs in all countries, including Chile [4, 5]. According to the World Health Organization (WHO), obesity is defined as an abnormal or excessive fat depots that presents a risk for health. The WHO recommends the use of body mass index (BMI) as a screening tool to identify individuals who are underweight, normal, overweight, or obesity. The BMI is commonly used in population-based surveys because it is a simple, low-cost index to classify nutritional status in adults, where a BMI equal or greater than 30 kg/m^2 is classified as obesity [6, 7]. BMI represents the sum of fat-mass index and fat-free mass index, but this index fails to reflect the distribution of adiposity/fat mass and fat free mass [8]. In middle-aged and older adults, body fat percentage is associated with risk of all-cause mortality regardless of the BMI [9].

Several methods for fat mass estimation are available, however, some of these are expensive and /or require trained health professionals to be applied [10]. For clinical and public health purposes it is necessary to have simple and accurate tools for estimating body fat percentage. Recently, Woolcott and Bergman [11] proposed an alternative to BMI named Relative Fat Mass (RFM), as a simple and low-cost estimator of whole-body fat percentage (BF%), using height and waist circumference. In a study in the US that included Mexican-Americans, European-Americans, African-Americans and Mexican adults, the RFM showed higher accuracy and precision than BMI to estimate BF % [12]. Moreover, RFM showed lower rates of total misclassification for obesity and higher diagnosis accuracy for type 2 diabetes patients (T2D) compared to BMI [11]. These positive findings require further confirmation in different populations to improve the understanding of the RFM as a new low-cost and non-specialized instrument to detect excess fat mass. Whether the RFM is a useful estimator of BF% in the Chilean population is unknown. Therefore, the aim of this study was to determine the accuracy of the RFM for 1) identifying individuals with elevated BF% and, 2) estimating the BF% comparing with dual-energy X-ray

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absorptiometry (DXA), as reference method, in the Chilean adult population as an alternative to BMI.

SUBJECTS AND METHODS

Participants

In this cross-sectional study, 270 healthy volunteers aged 18–65 years were recruited between October 2014 and June 2019, who had participated in several studies at the Department of Nutrition, Faculty of Medicine, University of Chile, as controls without pathologies except for obesity or hypertension. The studies were approved by the Ethics Committee of the Faculty of Medicine, University of Chile, and were performed in accordance with the 1975 Declaration of Helsinki and its revisions. Each volunteer signed an informed consent form for three test sessions in which DXA was measured, including a statement that the data would be kept secure and analyzed anonymously. Participants also agreed to be contacted for future secondary studies.

Anthropometric measurements

Waist circumference as well as body weight (kg) and height (m) were measured to calculate body mass index and RFM (BMI: weight in kilograms divided by the square of the height in meters, kg/m^2). Height was measured using a stadiometer SECA (model 220) and weight was measured using a calibrated digital scale SECA (model gmbh & co. Hammer Steindamm 3–25, 22089 Hamburg, Germany). Waist circumference (WC) was measured using a retractable tape (Urikjt™, Yiwu Miao Qin E-commerce Co., China) with a precision of 150 ± 1 mm. Waist circumference measurement was performed at the midpoint between the lower margin of the last palpable rib and the top of the iliac crest according to the WHO criteria [7]. BMI was classified according to the cut-off points defined by WHO [7].

Assessment of whole-body fat percentage (BF%)

The BF% was estimated by DXA. During the entire study, the same instrument (a Lunar Prodigy Advance equipment; General Electric Systems, Madison, WI), was used to determine FM (kg, and percentage), lean mass (LM, kg), and total bone mineral content (BMC, kg). The DXA equipment was calibrated with phantoms before each set of measurements. In our laboratory, the variability coefficient was 0.01 for FM and 0.40% for LM. As there is no universal consensus of a cut-off point for definition of obesity based on fat mass excess, we used the values proposed by Woolcott and Bergman, because of the aim of this study was to validate the RFM. Fat mass excess was defined as a BF% ≥22.8% for men and ≥33.9% for women. BF% measured by DXA was considered as a reference method [13], whereas RFM was calculated as follow:

64 – [20 × (height/waist circumference) both expressed in meters] + (12 × sex), being sex = 0 for men and 1 for women [11].

Statistical analysis

The normality assumption for the BMI, RFM and DXA was tested using the Shapiro-Wilk test. To analyze associations between BMI and RFM with DXA, the Pearson's correlation coefficient and the determination coefficient were calculated. To assess specificity and sensitivity a Receiving Operating Characteristics (ROC) curve was created comparing RFM and BMI with DXA. A test of equality comparing ROC areas between sex was applied. The Youden's Index (sensitivity/specificity-1) was calculated to identify the optimal cut off point for the BMI and RFM. Obesity and body fat excess percentage were considered according to the following cut off points: overweight was considered with BMI ≥25.0 kg/m^2, obesity with BMI ≥30.0 kg/m^2 and body fat excess percentage with ≥22.8% for men and ≥33.9% for women. Concordance between the BF% results measured by DXA and the RFM was assessed using the Bland–Altman Analysis. Slopes and intercepts were obtained by a regression analysis between the difference (response) and average (explanatory variable) of BF% measured by DXA and RFM. Statistical significance was set at $p < 0.05$. All the analyses were done with the software IBM SPSS Statistics v.25 and STATA SE v.16.

RESULTS

Characteristics of the participants

A total of 270 participants were included in the study, 46.3% ($n = 125$) were women and 53.7% ($n = 145$) men. Anthropometry and variables obtained by DXA in men and women are presented in Table 1. According to BMI, 27.0% ($n = 73$) of the participants had obesity, while according to DXA and RFM, 78.1% and 71.5% of the individuals had excess BF%, respectively. In addition, 28% ($n = 76$) of the participants had hypertension (data not shown). The Shapiro–Wilk test revealed that BMI, RFM and DXA were not normally distributed ($p < 0.05$).

Correlations analysis of reference vs. predicted body fat percentage

Pearson´s correlations coefficient between DXA, RFM and BMI were calculated for the total group and separately for women and men, with no significant differences between groups. Correlation coefficient between DXA and RFM for the total group was 0.88 ($p < 0.000$); and 0.88 ($p < 0.000$) and 0.81 ($p < 0.000$) for men and women respectively (Fig. 1A). Similarly, the correlation coefficient between DXA and BMI was 0.59 ($p < 0.000$) for the total group; 0.85 ($p < 0.000$) and 0.82 ($p < 0.000$) for men and women respectively (Fig. 1B). Additionally, the correlation coefficient for the total group between DXA and RFM (0.88) and DXA and BMI (0.59) was significantly different ($p < 0.000$).

Analysis of sensitivity and specificity

For the analysis of sensitivity and specificity, a Receiving Operating Characteristics (ROC) curve, area under the curve (AUC) and Youden's Index were calculated (Fig. 2). For men, when comparing RFM with DXA, the AUC of ROC curve was 0.970 (Fig. 2A), and for BMI with DXA the AUC of ROC curve was 0.959 (Fig. 2B). For women, the AUC of ROC curve comparing RFM with DXA was 0.946 (Fig. 2C), while the AUC of ROC curve comparing BMI with DXA was 0.942 (Fig. 2D). The analysis revealed no significant difference in the area under the ROC curve between RFM and BMI for men ($p = 0.420$) and women ($p = 0.750$). The cut off points obtained for men were 22.7% for RFM (Youden's index of 0.915),

Table 1. Characteristics of the 270 participants (mean values ± SDs).

	Women ($n = 125$)	Men ($n = 145$)	All ($n = 270$)
Age (years)	40.4 ± 16.3	47.8 ± 15.0	44.4 ± 16.1
Height (m)	1.6 ± 0.1	1.7 ± 0.1	1.6 ± 0.1
Weight (kg)	65.6 ± 13.4	81.5 ± 15.1	74.2 ± 16.4
WC (cm)	83.3 ± 14.3	95.0 ± 13.8	89.6 ± 15.2
WBF DXA (%)	38.2 ± 5.5	28.2 ± 8.3	32.6 ± 8.7
WBF RFM (%)	36.7 ± 6.9	27.1 ± 5.5	31.6 ± 7.8
BMI (kg/m^2)	26.0 ± 5.1	27.7 ± 4.7	26.9 ± 5.0
BMI category % (n)			
<18.5 (kg/m^2)	2.4 (3)	0 (0)	1.1 (3)
18.5–24.9 (kg/m^2)	48.8 (61)	29.7 (43)	38.5 (104)
25–29.9 (kg/m^2)	24 (30)	41.4 (60)	33.3 (90)
≥30,0 (kg/m^2)	24.8 (31)	29.9 (42)	27.0 (73)
BF% by DXA category % (n)			
Normal	20.0 (25)	23.4 (34)	21.9 (59)
Excess	80.0 (100)	76.6 (111)	78.1 (211)
BF% by RFM category % (n)			
Normal	31.2 (39)	26.2 (38)	28.5 (77)
Excess	68.8 (86)	73.8 (107)	71.5 (193)

WC waist circumference, *BF% DXA* body fat percentage measured by dual X-ray absorptiometry, *BF% RFM* body fat percentage calculated by Relative Fat Mass.

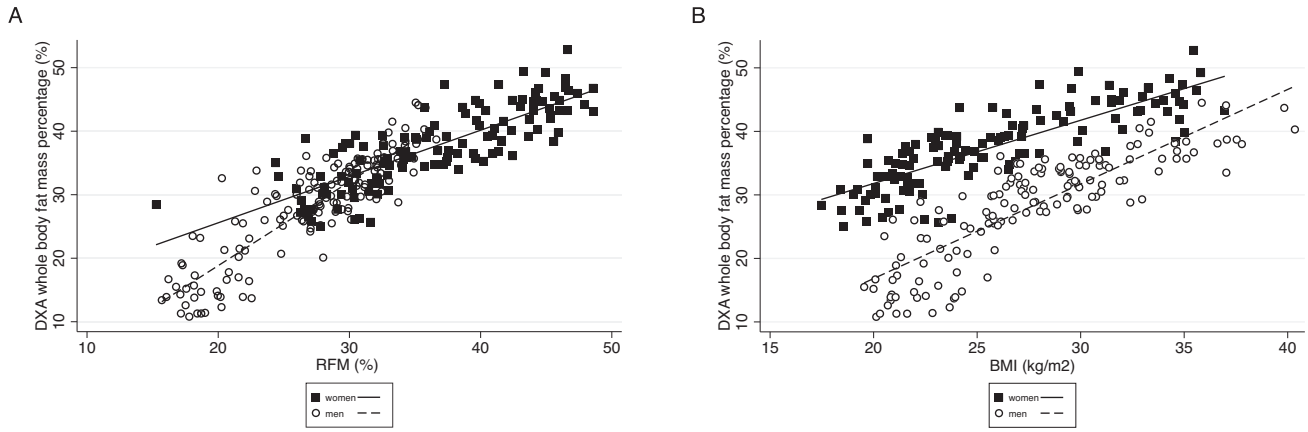

Fig. 1 Correlation of body fat percentage with RFM and BMI. Linear regression on the validation dataset was used to correlate body fat percentage with RFM (**A**) and BMI (**B**) in both sexes. BMI body mass index, DXA dual-energy X-ray absorptiometry, RFM relative fat mass.

and for BMI 24.4 kg/m^2 (Youden's index 0.853). Whereas the cut off points obtained for women were 32.4% for RFM (Youden's index 0.821), and 24.1 kg/m^2 for BMI (Youden's index 0.764).

Concordance analysis of reference vs. predicted body fat percentage

The difference between the DXA and RFM measurements (DXA-RFM) is normally distributed (Shapiro–Wilk test) in the total group ($p = 0.675$) as well as in women ($p = 0.350$) and men ($p = 0.059$). Mean difference (95% confidence intervals) of BF% between the DXA and RFM was 1.08 (0.58; 1.57) in the total group and 1.11 (0.42; 1.81) in men and 1.04 (0.32; 1.75) in women. The results of the linear regression between the differences and means of RFM and DXA after the Bland-Altman analysis is presented in Table 2. In the total group as well as in women and men, the coefficient analysis of the slope and the constant reveal that the bias (difference between DXA and RFM) is not constant (Fig. 3A). In men as the mean of the DXA and RFM measurements increases, the RFM tend to underestimate the measure of BF% (Fig. 3B). Conversely in women as the mean of the DXA and RFM measurements increases the RFM tend to overestimate the measure of BF% (Fig. 3C). However, the slope for women is closer to zero and less significant than for men. Similar results detected in men are present in the total group. Overall, these results suggest that the estimation bias is more pronounced in men than in women. The heteroskedasticity analysis revealed that the assumption constant variance of error terms is met in the total and both groups.

DISCUSSION

In this study, we evaluated whether RFM, developed as an index of body fat percentage (BF%) in a sample of US Americans [11] is useful as an alternative to BMI to identify individuals with elevated BF% and to estimate total body fat percentage in a population of Chilean adults. Our main results were as follows: 1) RFM is a reasonably accurate approach for identifying individuals with elevated BF%, and 2) our data show that there is a close agreement between BF% from RFM and DXA measurements, however, as the mean of the DXA and RFM measurements increases, RFM tends to underestimate BF% in men and overestimate BF% in women.

The equation proposed by Woolcott and Bergman included a representative US sample with a small percentage of Mexican-Americans (8.2%) [11], thus it may not be applicable to other ethnic groups such as ours. Therefore, it is highly relevant to validate the equation in specific groups such as Latin American population.

There are multiple methods of assessing body composition, more complex laboratory methods such as isotopic dilution, DXA, and air displacement plethysmography which are more accurate to assess body composition; however, they are expensive and require trained personnel. Other techniques such as skinfolds measurement and bioelectrical impedance are simpler than laboratory methods; however, require specific equipment and adequately trained personnel to use the equipment and to interpret the results of the measurements [10]. To overcome the limitations of more complex and expensive methods, Cichosz et al. [14] developed a new tool for assessing fat and lean body mass, BOMP (body composition prediction). The model, which includes sex, age, height, weight, waist circumference and triceps skinfold, had a correlation coefficient of R = 0.98 (SEE 2.07 kg) for predicting BF%. RFM has shown a better correlation with body fat mass than BMI against BF% estimated by the 4 compartments model [12], which is considered the gold standard for BF% [15]. Thus, RFM is more applicable to estimate the BF% and consequently to estimate the nutritional risk associated with overweight and obesity.

To our knowledge, there are a few studies which have evaluated the performance of the RFM against DXA in the Latin American adult population. In 81 healthy men aged 18–35 years from Brazil, the BF% measured by RFM showed a better correlation than BMI (0.90 vs. 0.79, respectively, $p < 0.05$), and a substantial agreement measured by Kappa coefficient (0.75 vs. 0.67) when compared to BF% measured by DXA as a reference method [16]. Similarly, in 61 healthy individuals (29 women and 32 men) aged 20–37 years from northern Mexico the RFM better predicted BF% measured by DXA [12]. Although, we showed similar results, comparisons are difficult considering that previous studies included a smaller sample size, only men [16] and, a small proportion of individuals with overweight and obesity [12].

Considering that, to date, there is no consensus in the cut-off point to assess body fat percentage, the most accepted values used as references were developed by Gallagher et al. [17] who established a range for BF% according to BMI, sex and age. The results by Gallagher are close to those defined by Woolcott and obtained in our study.

Our results support that the RFM is more accurate to identify excessive body fat in individuals without obesity according to BMI (Table 1) whereas BMI may lead to a misleading interpretation in the presence of either low or high fat free mass. Even if the BMI is high because of excessive body fat, it does not tell the clinician where the excess fat is located [18]. In addition, while new tools have the potential to simplify the assessment of BF%, they require to be validated in different populations and appropriate personnel training, making them challenging to use in large-scale studies.

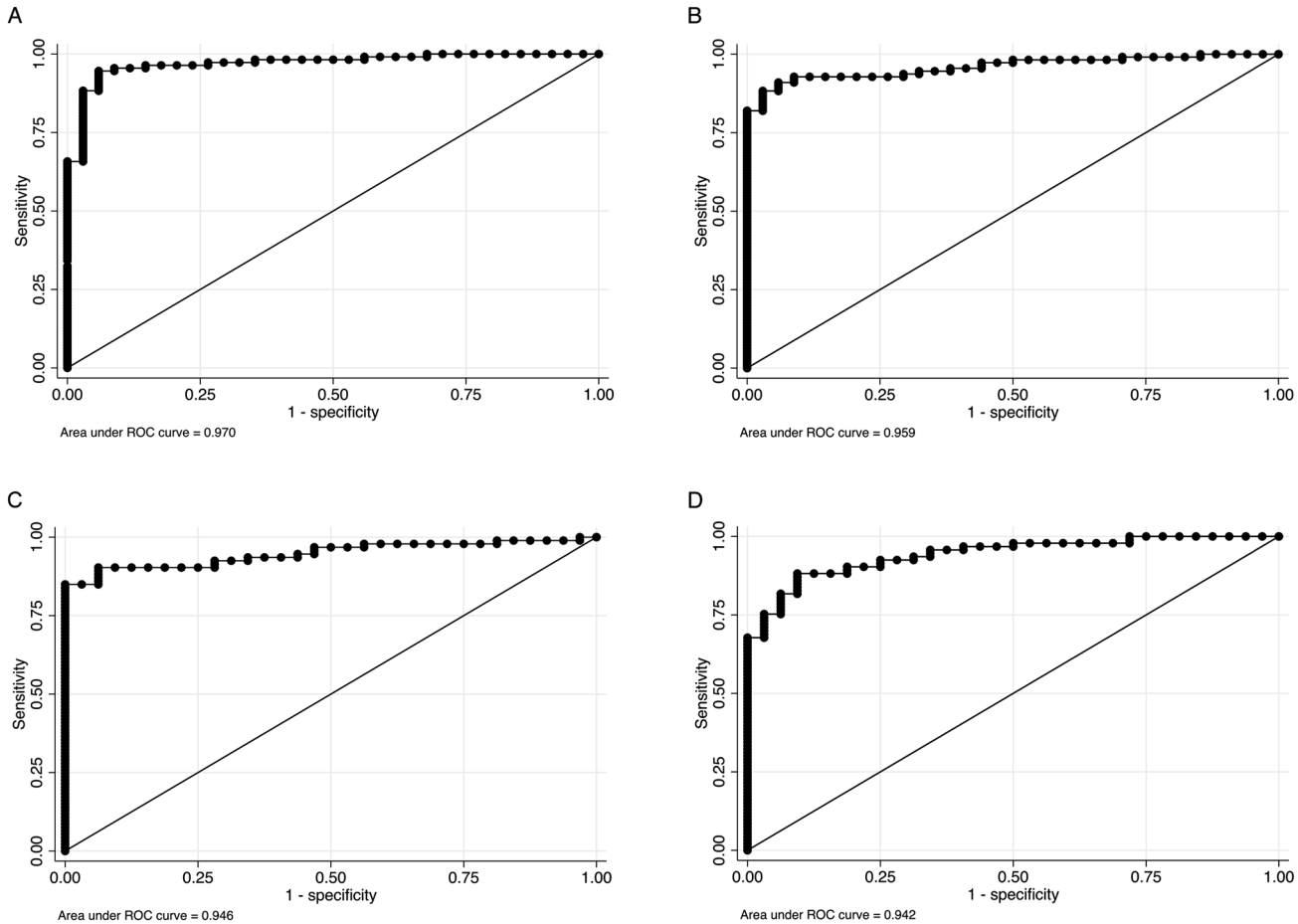

Fig. 2 Receiver operating characteristic curve, area under the curve and Youden's Index. Receiver operating characteristic curve, area under the curve and Youden's Index were calculated for RFM in men (**A**), BMI in men (**B**), RFM in women (**C**), and BMI in women (**D**). ROC receiver operating curve, AUC area under the curve, Youden's Index: sensitivity+specificity−1.

Table 2. Results of the linear regression analysis between the differences and means of RFM and DXA in the total group and separated by sex.

Total			
Predictors	**coefficient**	**95% CI**	***p*-value**
Constant	−2.78	−4.77; −0.79	0.006
Slope	0.120	0.06; 0.180	0.000
Breusch–Pagan/Cook–Weisberg test for heteroskedasticity, p = 0.695			
Men			
Constant	−10.45	−12.68; −8.22	0.000
Slope	0.417	0.33; 0.49	0.000
Breusch–Pagan/Cook–Weisberg test for heteroskedasticity, p = 0.157			
Women			
Constant	5.50	1.19; 9.82	0.013
Slope	−0.11	−0.23; −0.00	0.040

Breusch–Pagan/Cook–Weisberg test for heteroskedasticity, $p = 0.144$.

RFM can be easily calculated, using only two basic anthropometric measurements (height and waist circumference) that are usually obtained in a clinical medical and nutritional assessment using non-expensive equipment. This finding is relevant considering, that BF% is independently associated with higher morbimortality [9, 19]. Recently, the same authors who proposed the RFM algorithm suggested new RFM cut-offs to diagnose obesity and predict mortality risk in a representative sample of the United States adult population [20]. Similarly, Andreasson et al., reported a better predictive capacity of RFM over BMI for all-cause mortality and severe liver disease, even though RFM was not superior to waist circumference or waist-to-hip ratio [21] and Cichosz et al. reported that RFM predicted certain diseases as diabetes, high blood pressure and arthritis prevalence and hospitalization better compared with BMI or waist circumference [22]. Furthermore, Machado et al. found that RFM predicted dyslipidemia and hypertension better than BMI, although not for non-alcoholic fatty liver disease and insulin resistance [23], whereas Zwartkruis et al. reported that compared to BMI and waist circumference, the overall discriminative ability of RFM predicting incidence of atrial fibrillation, heart failure and coronary artery disease in the general population was at least similar in women and better in men [24].

Our study has some limitations. The study group included participants who were free of pathologies except for obesity and hypertension. In addition, the age range did not include individuals younger than 25 years or older than 62 years. As a result, our results cannot be extrapolated to people with other pathologies or to other age groups. There are several strengths that counteract this limitation, including the large sample size, the wide range of BMIs, the balanced number of men and women.

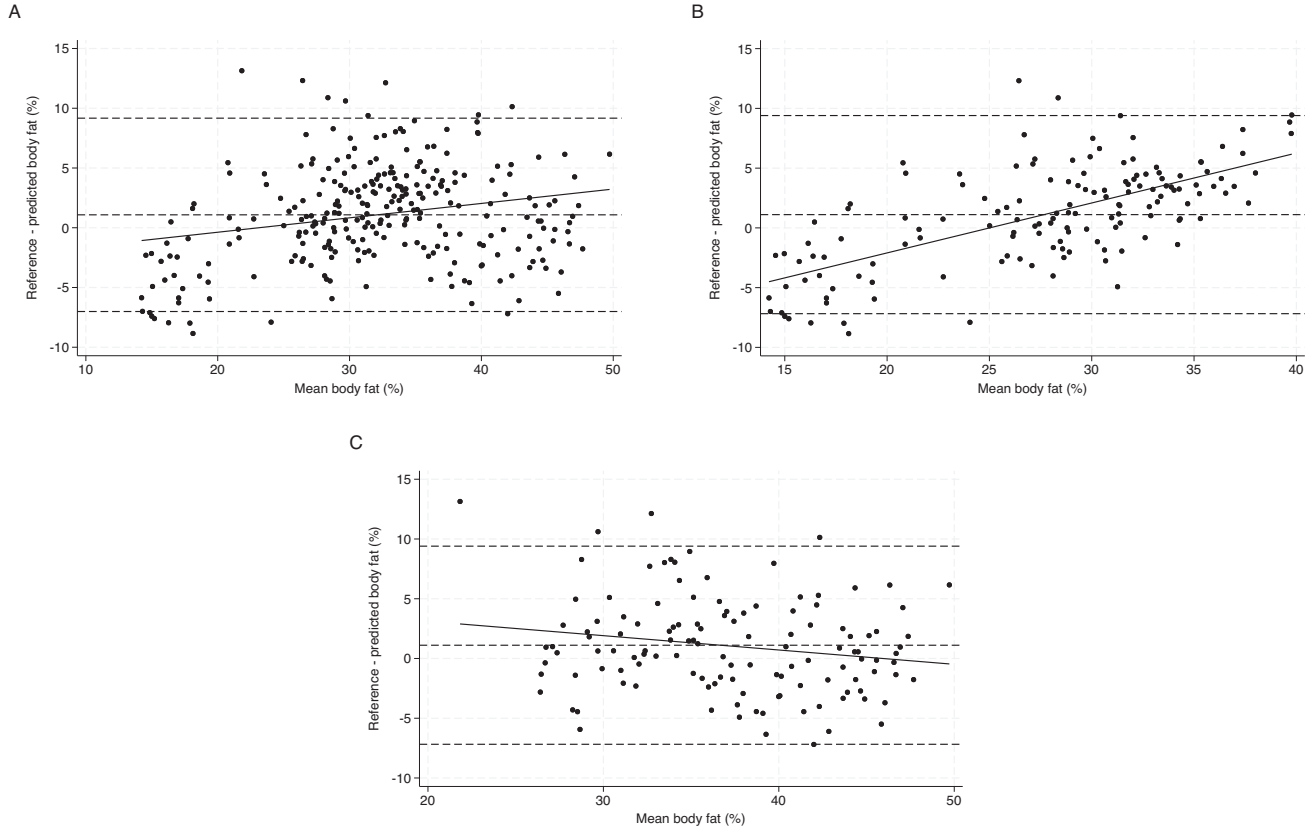

Fig. 3 Bland–Altman plot for comparison of reference and predicted body fat percentage. Reference body fat percentage was compared to the body fat percentage (BF%) estimated with the Relative Fat Mass (RFM) index in **A** the whole group, **B** men, and **C** women. Mean body fat percentage represents the average of reference and predicted body fat percentage.

Finally, although the currently accepted 'gold standard' is the four-compartment model [15] we use of DXA to validate RFM, which is considered a reference method for estimating BF%.

Based on our results, RFM is an accurate instrument to identify individuals with elevated BF% and to estimate BF%. However, the bias by sex must be considered when RFM is used to estimate the BF%.

This study supports that the RFM is an accurate alternative to assess the BF% in Chilean population. RFM showed a better correlation and diagnostic capacity for BF% compared to BMI. Although more studies are needed to investigate if the RFM is superior to existing alternatives in predicting comorbidities related with an excessive BF%, the use of RFM in clinical practice may help to assess health disturbances related to higher fat mass in the Chilean population.

DATA AVAILABILITY

Data from the study can be found within the article and additional data are available from the corresponding author on reasonable request.

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AUTHOR CONTRIBUTIONS

CA and PP were responsible for the study concept and design. MFT conducted the study. FC and JI contributed to data collection. AR, AO, MFT and PP were responsible for the analysis and data curation. All authors were involved in the assembly, analysis and interpretation of data and in drafting or revising the manuscript. All authors read and approved the final version of the manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

ETHICAL APPROVAL

The studies were approved by the Ethics Committee of the Medicine Faculty, Universidad de Chile, and were performed in accordance with the 1975 Declaration of Helsinki and its revisions.

ADDITIONAL INFORMATION

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